



BRAC UNIVERSITY

CSE461: Introduction to Robotics

Lab Project Report

Title: *[Home Security & Automation System]*
by

[Section-02]

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Abstract

This project aims to create a multifunctional home security system and automation system to improve the safety and also energy efficiency. In this robotic system we have used Arduino Uno and ESP32 controllers with different sensors including RFID, PIR, Raindrop sensor, temperature

sensor, flame sensor. RFID based door system enhance the home security, Flame sensor detect fire and send alert, PIR sensor reduce energy wastes by controlling fan, raindrop sensors active servo motor to retrieve cloths from being wet. The methodology enclose a modular architecture, the system is programmed in C++ via Arduino IDE and the assimilation of IOT for remote alert system. Final result establish a promising autonomous operation of all main functionalities including proving a effective security, sending fire alert and promote to sustainable use of energy. This home automation system robots impact mainly lies in cost effectiveness, scalable architecture, and high potential to improve safety systems in smart homes.

Keywords

[3 - 5 keywords, “microcontroller, automation, IoT, sustainability, Iot, Security System, sustainability ”]

1. Introduction:

The increasing requirements for home security and sustainable energy consumption has facilitated the implementation of home security systems. This project is motivated by the need for affordable home security with simple integrated hardware and software systems. The primary objective of the project is to create a home security system that integrates different actuators with multiple sensors to enhance the security, environmental monitoring, automated appliance control. In this system the scope comprise interfacing sensors and actuators with microcontrollers. A fundamental principle of robotics and computer interfacing is to create an intelligent system that is able to solve real problems by taking sensory inputs from the environment

2. Related Work/ Inspiration:

Home automation system is a common robotic system now a days but most often is it works in specific aspects like security door system These are

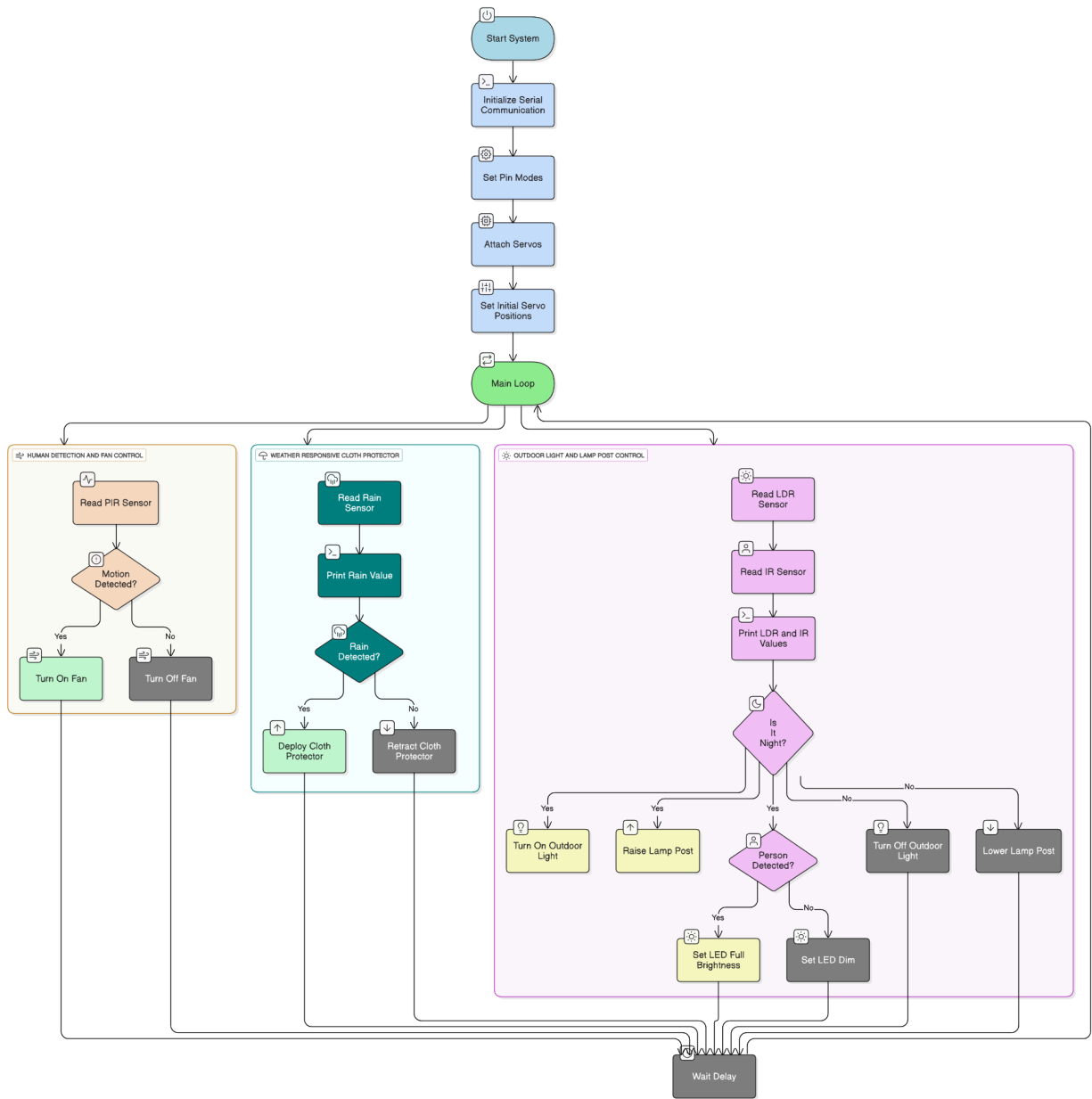
typically designed in specific domain like climate or security or lighting .Combining Smart door with camera module in commercial and control climate or lightning creates complex expensive product which is not affordable for middle class people that motivates me to work in this project and make this system in affordable price with more functionalities by combining fire detection,rain sensing,energy efficient human detection and light \fan controlling with less cost.

From existing projects our system is different in holistic integration and multi functional cohesion .Instead of using separated devices for multiple functions we have merged the system with diverse tasks in a synchronous way . By using Cross functional logic the system is able to make decision depending on the sensors combining the functionalities of specific sensors like PIR sensor works for both light control and fan activation instead of only light control and the speed of the fan can be controlled by the temperature sensor.Unlike fragmented solution our project creates a single integrated system that perform multiple tasks at a time

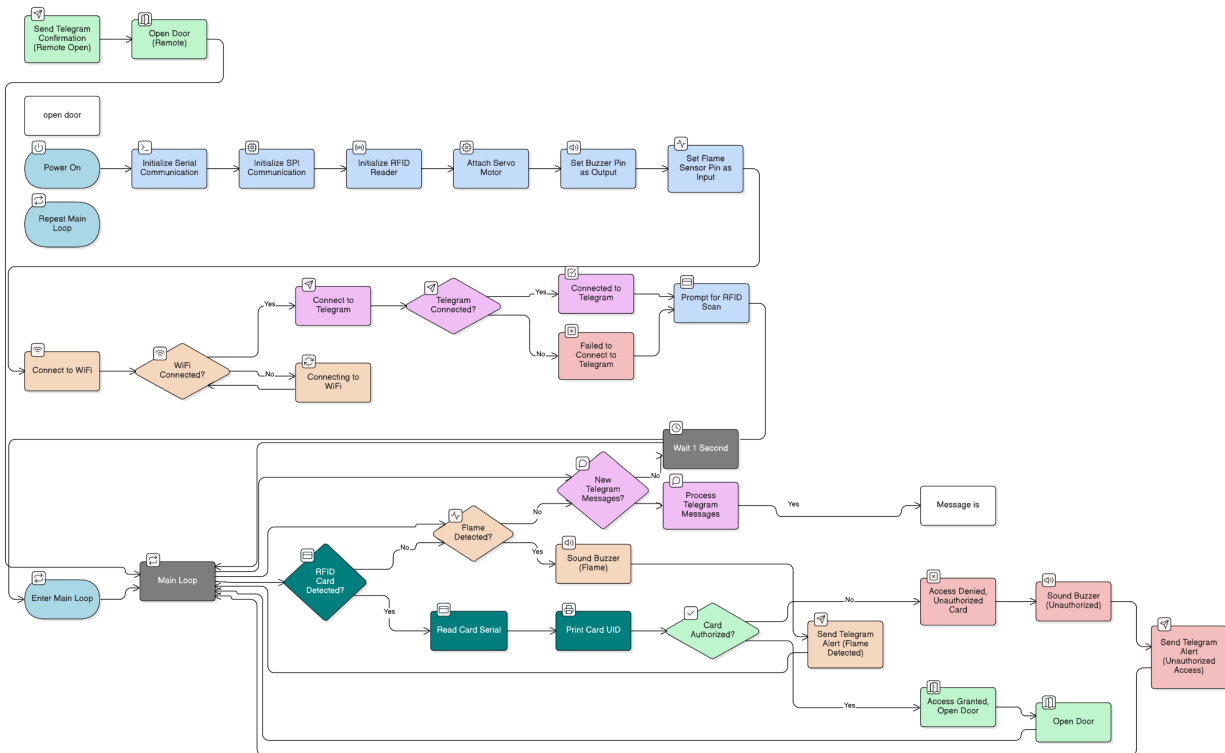
3. Technical Approach

☐ **System Architecture:** .

Workflow (Functionality 2,4,5)



Workflow(Functionality 1 and 3)



eraser

□ Components Used:

Hardware

- **Arduino Uno:** Arduino Uno is the main microcontroller of our project that is used to control the actuators and the sensor. It works based on the input signals and after processing the output signals send to the motors
- **ESP32 cam Module:** Another controller that is working on wifi and camera for the alert system, sending pictures to the mobile through the wifi protocols when fire is detected and an unauthorized person wants to enter.
- **Servo Motor:** Servo motor works on PWM method, It is used to rotate in an angle. It Controls door lock & cloth rod movement.

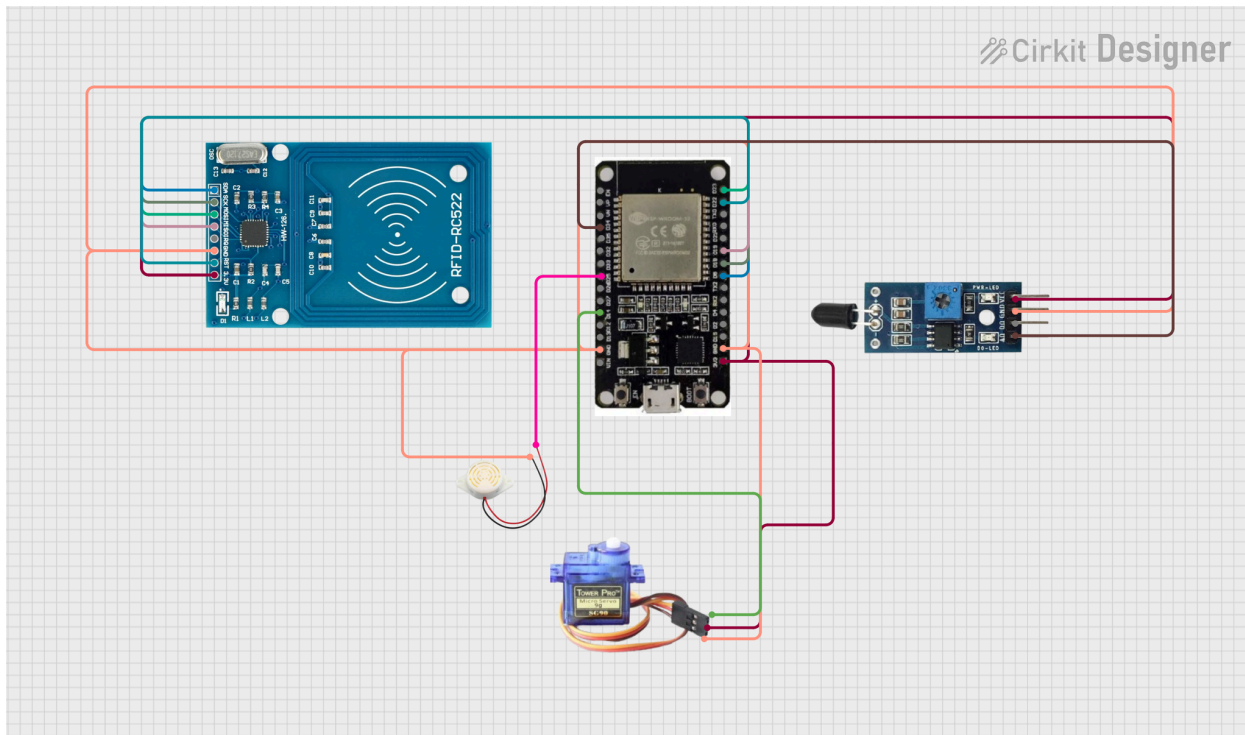
- DC Motor: DC motor mainly converts electrical energy into mechanical rotation which is used in the fan system, It activates the fan from getting the signal from PIR and temperature sensors. Here DC motor is controlled by motor driver L298N
- RFID Sensor: Used for door access authentication. It is used radio frequency.
- PIR Sensor: Used to detect human movement by infrared radiation emitted from human body and control fan light activation system
- Flame Sensor: Detects fire hazards and IR light increase and turns the buzzer on and triggers ESP alert .
- Temperature Sensor: Measure room temperature and convert them into electrical signals, Arduino compares the input signal with the threshold and activates the fan .
- Raindrop Sensor: sense the rainfall and take steps for cloth protection.
- LDR Sensor: Measures ambient light ,In dark places, the resistance is high which creates low output voltage and in bright the resistance is low, which creates the high output voltage for smart lighting.
- Buzzer: Arduino controls the buzzer by sending high signal to activate the alerts on unauthorized access and fire.
- Breadboard, Jumper Wires, Resistors, Battery, multimeter etc are the some basic components of the projects.

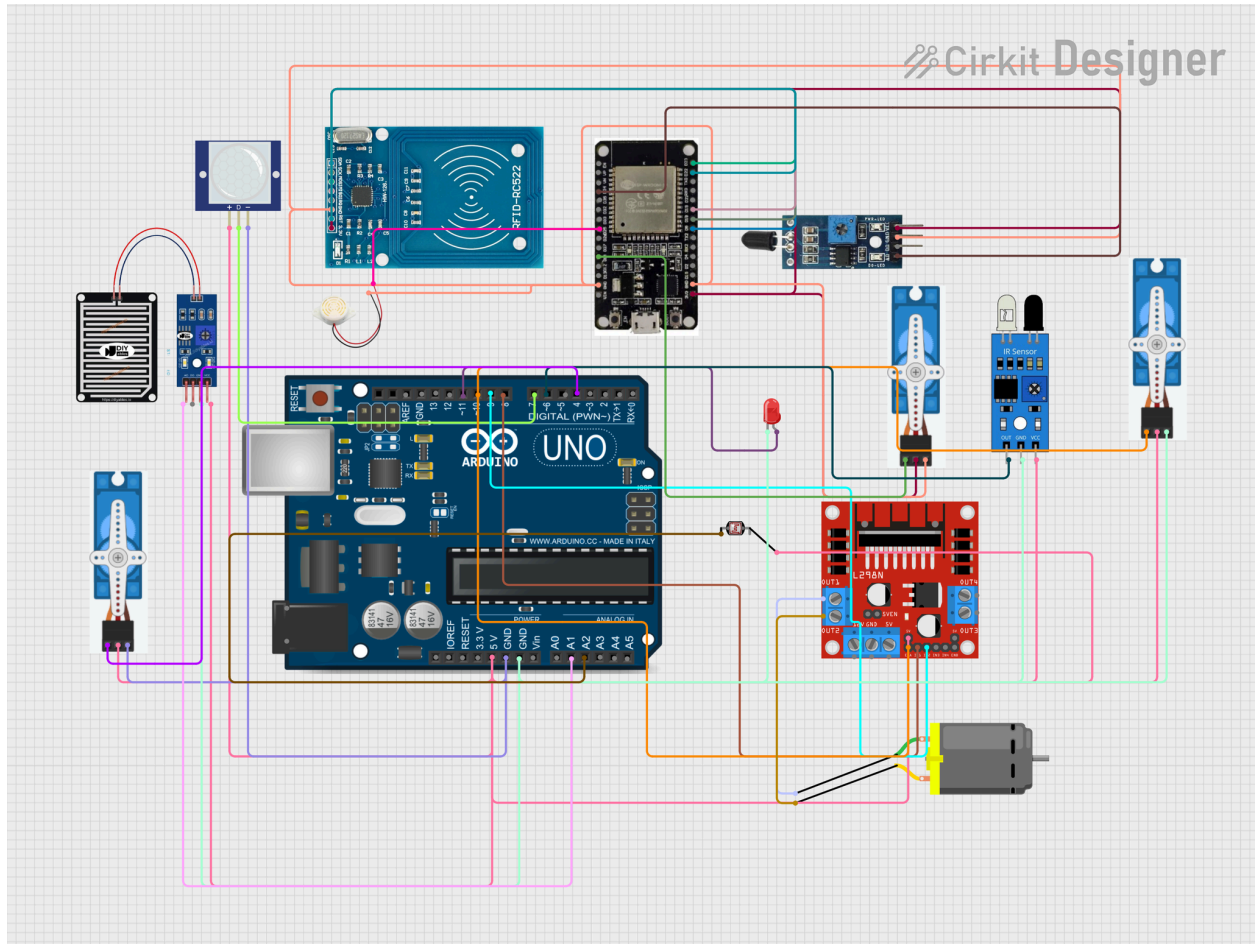
Software

- Embedded C / Arduino C++: Programming language.
- ESP32 cam Libraries (Wi-Fi, Blynk): For notification system.
- Tinkercad: For circuit design & simulation.



Circuit/Schematic: Provide diagrams.





□ **Functionality:**

In our home security system project ,we are working on 5 functionality.The working procedure is given below.

Step 1:The RFID sensor scans the RFID card and if the card is authenticated the servo motor will open the door for the person .If not ,an alert will be sent to the owner of the house and if the owner wants to enter the person ,the owner will send “open door” message and servo will open the door .

Step2: After entering the house ,the PIR sensor detect the human and turn the fan on by using servo motor and if the person leave the place the fan will again turn off

Step3-If any unwanted fire occurs in the house,flame sensor detects the fire and buzzer will be activated and also a message will be sent to the registered number by esp module.

Step4-When rainfall starts,the rain sensor detects the rain and the servo motor rotates the cloth stick and protects the cloth from getting wet.

Step5-By using LDR sensors ambient light level is observed and when it's dark the system turns on the light and also if a human is detected the brightness of the light will increase.

Challenges:We have faced different types of challenges to make this project

Technical Challenges:

- 1.Component Change- We decided to use an ESP32 cam but it does not have enough ports so we decided to shift to ESP32.
- 2.Sensors Regulation-Different sensors has different sensitivity,it was difficult to merge them after testing
- 3.Power supply Issue-Arduino operates in 5V but ESP32 operates in 3.3v.These different voltage ranges also created issues.

Integration challenges:

- 1.Power failure-TThe charge of the 9v battery was finished multiple times which creates issue when using Dc motor
- 2.Temperature sensor fluctuation-It was very difficult to have a proper signal from the temperature sensor because of circuit noise.

Budget Limitation:We are supposed to do an impactful robotic system at minimum cost.So we had to minimize many components like extra batteries,motors,breadboard etc.

4. Sustainability & Impact

- ☐ **Sustainability:** The project focuses on energy efficiency by automatically controlling the fan and light when no human is available in the room, thus saving electricity. Using a DC motor is more efficient than using an AC fan. In this project, multiple components are used which are reusable.
- ☐ **Impact:** This project has a strong impact on users for increasing safety and providing an immediate alert after detecting fire, which makes the user more reliable on this system. Moreover, automating tasks helps the elderly people. Finally, its affordable price makes it possible for people to achieve this system without hampering their daily life.
- ☐ **Future Work:** This project can be more functional in the future by including a mobile app for control, solar power integration, adding more sensors, and including machine learning models on the sensor to predict behavior.
- ☐ **Limitations:** The short range of RFID, setup of Arduino Uno which has some restrictions in adding components, limited Wi-Fi range of ESP32 are some limitations of this system that were observed despite the successful implementation of the project.

5. Results & Discussion

Testing outcomes:

1 RFID sensor

Test	Expected Outcome	Actual Outcome	Result

1.Authorized card	Servo open the door	Door open Successfully	Correct
2.Unauthorized card	RFID send alert	Send alert successfully -Remaining the door close(No message form the owner)	Correct
3,Unauthorized card	RFID send alert	Send alert successfully -Open the door(If "Open Door" message is send from the owner)	Correct
3. NO card	No action	No action	Correct

2.Flame sensor

Test	Expected outcome	Actual outcome	duration
1.Flame present	Flame detect and send alert	Flame detect and send alert	5s
2.No flame	No response	No response	

3.Rain sensor

Test	Expected outcome	Actual Outcome	Result
1.Rain	Servo rotates	Servo rotates	Protect cloth
2.No rain	No response	No response	Keep as it is

4.PIR sensor

Test	Expected output	Actual output
Motion detect	Servo control fan	Servo response
No motion detect	No response	No reason

Result : All the functionalities performed perfectly.Unauthorized card is detected and send alert,fire alert is send , fan control is working, light automation is controlled,and rain drop sensor is also responding.Overall the project is reliable for people and it takes vary less time to response .The efficiency of this project is high for giving appropriate result for every sensors .

Discussion: The home security and automation system merged multiple sensors and actuators for creating a smart home.This system confirms the authorized locked system,fire alert system,light and fan control and responsive to the weather.The fast response, make the project more effective.The overall system ensure to use less energy.

6. Conclusion

This project prosperously implemented a cost effective,energy efficient, integrated Home security and automation system.The key achievements of this project is accumulation of different actuators with multiple sensors for establishing a system that manages access control,risk detection,climate control etc.This system proves a small computer can be used to create a safe home using energy efficient tools with less cost.The outcomes ensures that it is a economical, practical ,efficient,effective embedded system that protects dwelling,save energy ,enrich daily activities .

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